Axial Reorganization and Group Offset Mamba for Hyperspectral Image Classification

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***Abstract*—Hyperspectral image classification (HSIC) is a core task in the field of remote sensing. Its inherent characteristics of high spectral redundancy, complex spatial structure, and strong heterogeneity of land cover make it difficult to capture spatial global correlations and inefficiently model spectral long-range dependencies. Traditional methods are difficult to balance feature discriminability and modeling efficiency, and the accuracy of fine classification in complex scenes is limited. In this article, an axial reorganization and group offset Mamba (ARGO-Mamba) network is proposed. First, a local–global attention guidance (LGAG) module is proposed, which preserves key fine information through local feature aggregation and then dynamically enhances the representation of important features during the discriminative feature calibration process. Second, an axial reorganization spatial Mamba (ARSM) module is introduced, which abandons the optimization of scanning strategies in spatial modeling and directly encodes the directional structural information of land features through axial reorganization and captures global spatial contextual information more efficiently. Finally, a group offset spectral Mamba (GOSM) module is constructed, which creates diverse spectral grouping views through spectral grouping and group shift and simultaneously models its long-range spectral dependencies using Mamba. Experiments on the Indian Pines, Pavia University (PU), Houston 2013, and WHU-Hi-Longkou datasets show that the OA values of the proposed ARGO-Mamba increase by 0.56%–2.5% compared to the most advanced methods in recent years, with an average increase of 1.3%. It performs more outstandingly in small sample situations and highly heterogeneous scenes, providing a new solution for complex remote sensing classification tasks. The source code will be released on https://github.com/liang-zhenlong/ARGO-Mamba**

***Index Terms*—Attention mechanism, group offset, hyperspectral image classification (HSIC), Mamba, spectral–spatial model.**

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I. Introduction

REMOTE sensing technology can efficiently obtain surface observation data with high spatiotemporal resolution. Among them, hyperspectral images (HSIs) generate nearly continuous spectral curves for each pixel by imaging on hundreds of consecutive narrow bands. Compared with traditional RGB images that only contain three broad bands of red, green, and blue, HSI has stronger discriminative ability in the spectral dimension, which can significantly improve the accuracy of land cover recognition and classification. Thanks to this advantage, HSI has shown broad prospects in various remote sensing application scenarios such as urban planning [1], [2], military reconnaissance [3], [4], precision agriculture [5], [6], and so on. In these tasks, the goal of hyperspectral image classification (HSIC) is to assign a corresponding category label to each pixel in the image [7].

In early HSIC research, most methods relied on manually designed feature descriptors and subspace learning strategies [8], [9], [10], [11]. In recent years, deep learning has become the mainstream paradigm in the field of HSIC. Early attempts mainly focused on unsupervised learning [12], [13]. Subsequently, convolutional neural networks (CNNs) were widely adopted [14], [15], [16], [17]. In order to further explore the spatial spectral characteristics of hyperspectral data, researchers have proposed various improved architectures, such as feature learning based on tensor deep autoencoders, deep ring block networks, and graph neural networks combined with diffusion models [18], [19], [20]. By leveraging weight sharing and local connections, convolution operations can efficiently capture spatial local features of images. However, CNNs are limited by local receptive fields and struggle to effectively capture global contextual relation-

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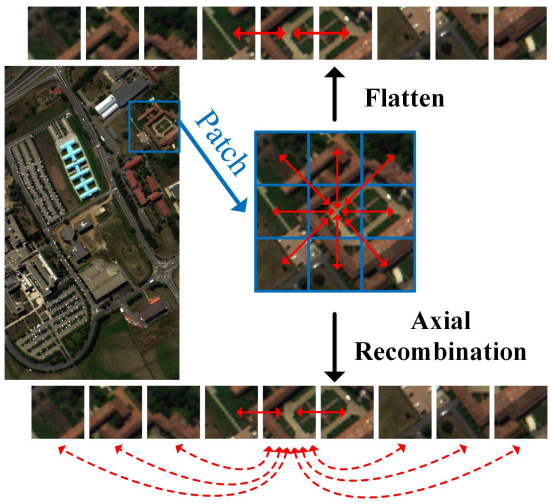


Fig. 1. Schematic of research motivation. In local spatial features, there is a significant semantic association between the central pixel and its neighboring pixels (indicated by the red solid line). If the 2-D spatial features are flattened directly into a 1-D sequence, the original semantic associations will inevitably be lost. To alleviate this problem, this article proposes an axial restructuring strategy, which preserves 2-D semantic relationships (indicated by the red dashed line) in 1-D sequences by prestructuring them along the axial direction, thereby minimizing semantic loss to the greatest extent possible.

ships, resulting in limited performance when dealing with complex-shaped terrain. Later, Transformers brought revolutionary changes to HSIC with their self-attention mechanism [21], [22], [23]. Transformer can effectively model global context and long-range dependencies, significantly overcoming the local limitations of CNNs. However, the computational complexity of the self-attention mechanism is the square of the sequence length($O(n^2)$), which poses significant computational challenges and memory consumption, greatly limiting the practicality and scalability of Transformers in handling HSI.

Recently, the state space model (SSM) [24] has gradually become an efficient structure for constructing deep networks. Among them, the structured state space sequence structure (S4) [25] has been widely used in continuous sequence analysis due to its outstanding performance in long sequence modeling. On this basis, Mamba [26] further improved S4. For example, Mamba introduced hardware awareness in its design, enabling it to maintain strong long-range dependency modeling capabilities while achieving higher computational efficiency than Transformers [27]. With this advantage, Mamba has been widely applied in multiple research fields, including natural language processing (NLP) [25] and computer vision (CV) [28]. The preliminary applications in HSIC [29], [30], [31] also validated the potential of Mamba. However, due to the unique spatial structure and spectral characteristics of hyperspectral data, there are still several challenges that need to be addressed in effectively applying the sequence modeling paradigm to HSIC.

1) There is significant semantic correlation between land features in HSIs (see Fig. 1). Adjacent pixels usually have strong consistency in spatial position and semantic category, and their spatial contextual information is crucial for classification performance. However, existing methods often require restructuring or linearization of 2-D structures when modeling spatial features, thereby disrupting the original spatial topological relationships and local semantic consistency, limiting the effective mining of fine spatial discriminative information.
2) HSIs contain a large number of continuous and highly correlated spectral bands. Directly modeling all bands will incur high computational overhead, while common spectral grouping or compression strategies, if not designed properly, can easily disrupt the local continuity and correlation structure between key bands, thereby weakening the ability to characterize fine-grained spectral discrimination information. Therefore, how to maintain spectral discriminability while reducing computational complexity remains an important challenge in HSIC.

To this end, this article proposes the axial reorganization and group offset Mamba (ARGO-Mamba) network, designed to efficiently model long-range dependencies while maintaining spatial–spectral consistency. The framework jointly represents spatial and spectral features through a sequential modeling paradigm, addressing spatial semantic loss and spectral redundancy issues via local–global guidance and customized modeling strategies. This approach enhances classification performance and model robustness while maintaining high computational efficiency.

The main contributions of this article can be summarized as follows.

1) A local–global attention guidance (LGAG) mechanism is constructed, which can effectively alleviate the problem of excessive smoothing of local discriminative details by global statistics, thereby improving the discriminative and stable feature representation.
2) An axial reorganization spatial Mamba (ARSM) module is designed, which enhances the modeling ability of directionality and spatial topology consistency through structured spatial reorganization, enabling efficient sequence modeling paradigms to better adapt to complex spatial structure scenarios.
3) A group offset spectral Mamba (GOSM) module is constructed, which effectively alleviates the limitation of a fixed grouping strategy to key spectral continuity and related structures through multiview spectral grouping representation and achieves efficient modeling of long-range spectral dependencies.
4) A unified modeling framework ARGO-Mamba for HSIC is proposed, which synergistically models spatial structure consistency and spectral long-range dependence in the efficient sequence modeling paradigm, significantly improving the discriminative ability of hyperspectral features while ensuring computational efficiency.

The rest of this article is organized as follows. Section II reviews relevant work, Section III provides a comprehensive description of the proposed method, Section IV outlines experimental results and analysis, and Section V discusses conclusions and future work.

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II. Related Work

A. Progress of HSIC

1) CNN-Based Methods: CNNs became the mainstream approach for HSIC due to their ability to automatically learn features. For instance, Hu et al. [15] employed 1-D CNNs for feature extraction along the spectral dimension. However, HSI is prone to external interference during imaging, and spectral information among different land cover types often exhibits significant overlap, making it challenging to achieve ideal performance relying solely on spectral data. To address this, Lee and Kwon [16] proposed a contextual deep CNN to jointly model both spectral and spatial information. Early CNN methods mostly relied on direct stacking of convolutional layers to improve classification performance, but this approach often resulted in overfitting of the network. In response to this issue, researchers have proposed various improved architectures, such as PyResNet [32] using residual connections to alleviate overfitting, and the FDSSC network proposed in [33] achieving more efficient feature reuse through dense connections, thereby improving classification performance. However, the inherent local receptive field of CNNs still poses challenges in modeling global dependencies.

2) Transformer-Based Methods: Benefiting from the advantages of attention mechanisms in long-range sequence modeling and global contextual feature extraction, the Transformer architecture has been widely studied and applied to HSIC tasks. For example, SpectralFormer proposed by Hong et al. [22] treats the spectral sequence of a single pixel as a 1-D input and utilizes a self-attention mechanism to capture long-range dependencies between different bands. Based on this, HIT [35] integrates convolution operations to balance long-range modeling capabilities and local spatial–spectral feature representation. SSFTT [36] implements spatial–spectral feature tokenization through convolution and then introduces a Transformer for global modeling, thereby combining the advantages of CNN and Transformer. In addition, the A^2S^2K-ResNet [37] models spectral and spatial features separately and enhances feature expression ability by adaptively adjusting the size of the convolution kernel. Overall, although Transformer has demonstrated excellent performance in HSIC, its inherent self-attention mechanism leads to a quadratic increase in computational complexity with input length, posing significant challenges in modeling efficiency and memory overhead, thereby limiting its ability to model long-distance dependencies to some extent [27].

3) Spatial and Spectral Attention Methods: Given the presence of redundant information in hyperspectral data, spatial and spectral attention mechanisms are introduced to adaptively learn the importance of features. The spatial or spectral attention modules designed in existing related studies (such as [38], [39], [40]) all use global pooling operations to directly generate attention weights. However, this direct reliance on a global pooling scheme has limitations: due to its indiscriminate compression of global features, local key structures (such as terrain edges, fine textures) or local spectral-specific information are easily diluted and ignored, which affects the attention mechanism's ability to accurately focus on core discriminative features.

B. Progress of Mamba

Benefiting from its efficiency in sequence modeling [42], recent research has gradually extended Mamba to feature modeling in visual tasks. In the field of HSIC, various Mamba-based methods have emerged. For example, Zhu et al. [28] proposed ViMamba, which captures global spatial dependencies by bidirectionally scanning spatial features and flattening them into sequences. SpectralMamba [30] focuses on spectral dimension modeling and integrates local spatial information to enhance representation ability. Based on this, MambaHSI [31] achieves joint spatial–spectral modeling by combining spatial flattening with spectral grouping. On the other hand, VMamba proposed by Liu et al. [44] adopts a multipath 2-D scanning strategy to generate multiple sequences in both horizontal and vertical directions, thereby mitigating the information loss caused by spatial flattening. Huang et al. [45] further proposed Local-VMamba, which enhances local feature modeling while considering global perception through windowed selective scanning and cross-window interaction. In addition, MambaLG [46] combines spatial flattening with a sliding subband grouping mechanism to achieve efficient modeling of global spatial–spectral features. However, existing Mamba methods are generally limited to adjustments to scanning strategies and ignore the effective expression of local continuous spectral structures, making it difficult to fully utilize Mamba's modeling potential on hyperspectral data.

III. Methodology

This section first reviews the development of SSMs and focuses on the process of evolution into Mamba. Subsequently, the overall framework of the proposed ARGO-Mamba model is introduced. The ARGO-Mamba architecture comprises three key modules: LGAG, ARSM, and GOSM. Next, we will provide a detailed explanation of each module in sequence.

A. Preliminary Information

SSM is based on the principle of linear time-invariant (LTI) systems, which map continuous input signals $x(t) \in \mathbb{R}^L$ to continuous output signals $y(t) \in \mathbb{R}^L$. In particular, it achieves this mapping through a latent state $h(t) \in \mathbb{R}^N$. This process can be expressed using an ordinary differential equation (ODE) as follows:

$$\begin{cases} h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) = \mathbf{C}h(t) + \mathbf{D}x(\mathrm{t}) \end{cases} \tag{1}$$

where $\mathbf{A} \in \mathrm{R}^{N\times N}$ is the state matrix, which describes the changes in the internal state of the system. $\mathbf{B} \in \mathbb{R}^{N\times L}$ is the input matrix, $\mathbf{C} \in \mathbb{R}^{L\times N}$ is the output matrix, and $\mathbf{D} \in \mathbb{R}^{L\times L}$ is the direct mapping matrix. $h'(t) \in \mathbb{R}^{\mathrm{N}}$ represents the rate of change of the system's state at time point t.

In order to extend the continuous-time system in (1) to discrete sequence modeling tasks, the S4 framework discretizes it. In particular, S4 uses a zero-order hold (ZOH) technique with a time scale parameter $\mathbf{\Delta} \in \mathbb{R}^L$ to map parameters $\mathbf{A}$

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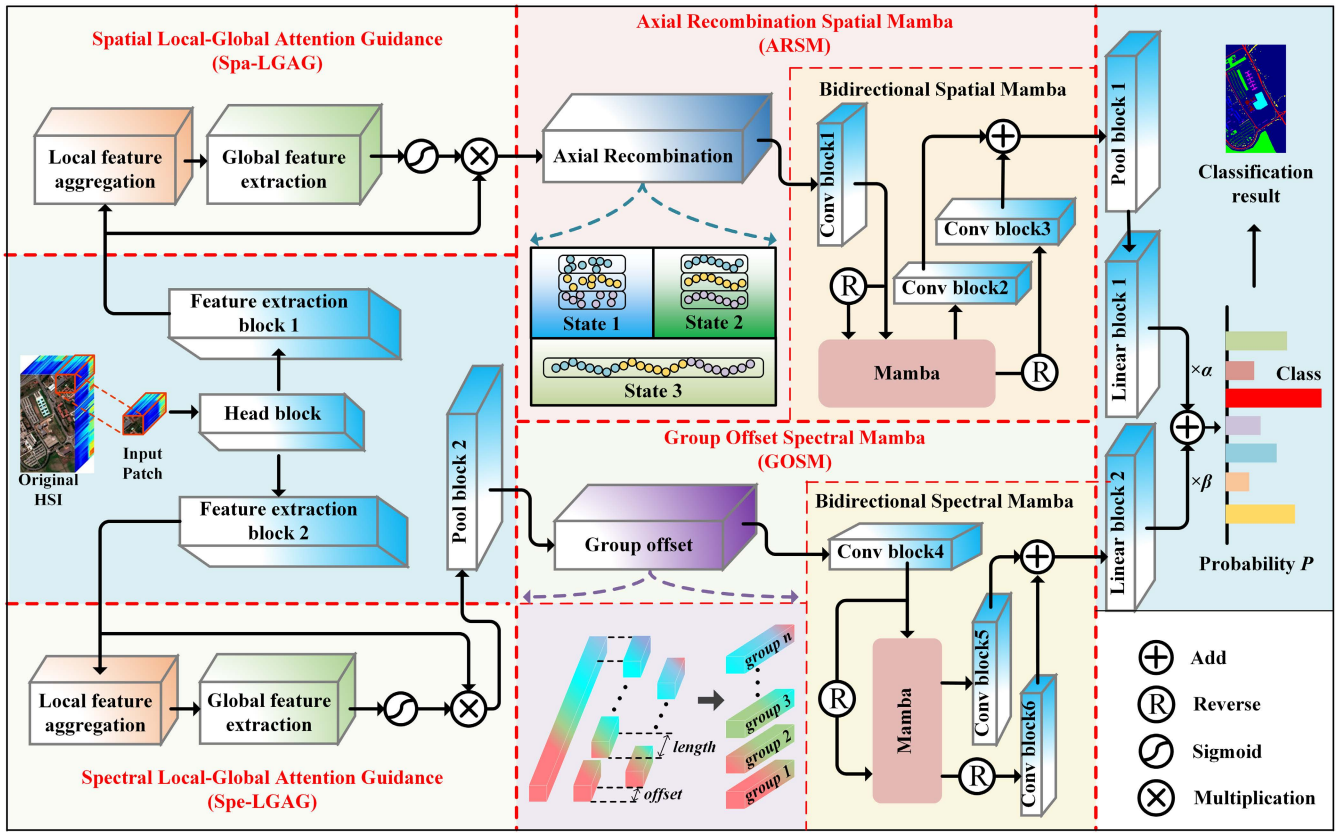


Fig. 2. Overall structure of the proposed ARGO-Mamba network. Among them, the head block is used to adjust the number of channels. Feature extraction blocks 1 and 2 are used to extract local spatial and spectral features, respectively. Conv blocks (1–6) are used to further integrate information between channels. Pool blocks 1 and 2 are both used for dimension compression.

and $\mathbf{B}$ in a continuous system to parameters $\overline{\mathbf{A}} \in \mathbb{R}^{N\times N}$ and $\overline{\mathbf{B}} \in \mathbb{R}^{N\times N}$ in a discrete system. This process can be expressed as follows:

$$\begin{cases} \bar{\mathbf{A}} = \exp(\Delta\mathbf{A}) \\ \bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right) \cdot \Delta\mathbf{B} \end{cases} \tag{2}$$

where $\exp(\cdot)$ represents matrix exponential operation and $\mathbf{I}$ is the identity matrix. After completing the above discretization steps, the discretized SSM system can be further expressed as follows:

$$\begin{cases} h_k = \bar{\mathbf{A}}h_{k-1} + \bar{\mathbf{B}}x_k \\ y_k = \mathbf{C}h_k + \mathbf{D}x_k. \end{cases} \tag{3}$$

In order to enable the model to dynamically adjust based on the input sequence, Mamba introduced a selective mechanism in the S6 architecture. The core idea of this mechanism is to make parameters $\mathbf{B}, \mathbf{C}, and\mathbf{\Delta}$ no longer static but dynamically generated based on the input sequence x_k. In particular, for each input x_k, Mamba predicts its corresponding $\mathbf{B}_k, \mathbf{C}_k, \mathbf{\Delta}_k$ through a linear projection layer.

The discretization variables $\overline{\mathbf{A}}_k$ and $\overline{\mathbf{B}}_k$ can be further obtained through (2). At this point, the S6 architecture can be described as follows:

$$\begin{cases} h_k = \overline{\mathbf{A}}_k h_{k-1} + \overline{\mathbf{B}}_k x_k \\ y_k = \mathbf{C}_k h_k + \mathbf{D}x_k. \end{cases} \tag{4}$$

B. Overall Structure

Assuming the original HSI $\mathbf{I} \in \mathbb{R}^{H\times W\times B}$, where H and W represent the height and width of the image, and B represents the number of bands. Considering the spatial correlation between adjacent pixels, the image is divided into a series of 3-D patch cubes $\{\mathbf{X} \in \mathbb{R}^{B\times S\times S}\}_{i=1}^{H\times W}$. Among them, $S \times S$ represents the spatial dimension size of the patches. Subsequently, using the central pixel labels of each patch as supervisory signals, these cubes are used as inputs to the model.

The overall structure of ARGO-Mamba proposed in this article is shown in Fig. 2. First, to adapt the model to different datasets, the input data are processed through an initial convolutional layer to uniformly adjust the number of channels to C. Subsequently, the framework unfolds into a spatial–spectral dual-branch structure to synergistically leverage the complementary advantages of hyperspectral data. The spatial branch integrates the Spa-LGAG and ARSM modules, aiming to enhance key spatial features and model global context; the spectral branch combines the Spe-LGAG and GOSM modules to highlight important bands and capture long-range spectral dependencies.

In the final classification stage, the output $\mathbf{X}_{spa_out} \in \mathbb{R}^C$ of the spatial branch and the output $\mathbf{X}_{spe_out} \in \mathbb{R}^C$ of the spectral branch are dynamically fused to generate the final prediction result, which can be expressed as follows:

$$Prob = \alpha \cdot \text{Linear}_1\left(\mathbf{X}_{spa_out}\right) + \beta \cdot \text{Linear}_2\left(\mathbf{X}_{spe_out}\right) \tag{5}$$

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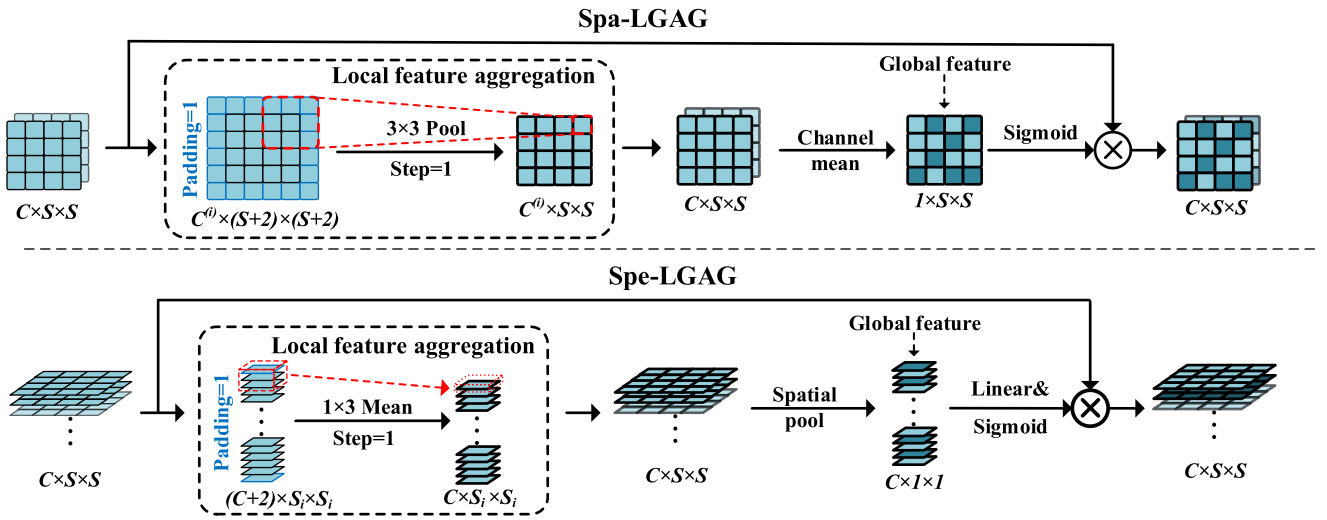


Fig. 3. Detailed structure of LGAG.

where *Prob* represents the final predicted probability distribution, and Linear(·) is the linear mapping operation. The coefficients α and β are obtained by applying a softmax to two learnable parameters w_1 and w_2, i.e., $[\alpha, \beta] = \text{Softmax}([w_1, w_2])$, so that $\alpha + \beta = 1$. This enables the model to learn the optimal contribution ratio between the spatial and spectral branches from data, rather than relying on fixed scalars.

C. Local–Global Attention Guidance

Existing attention frameworks typically rely on global pooling to extract global contextual information and weight features accordingly (such as CBAM [47]). However, this global statistics-based approach often ignores local feature information, which may lead to insufficient expression of fine spatial and spectral features. To overcome the above limitations, this article proposes an LGAG module. Considering the differences between spatial and spectral features, Spa-LGAG with spatial branches and Spe-LGAG with spectral branches were designed, and their structures are shown in Fig. 3.

1) Spa-LGAG: To enhance the expression of fine spatial features, Spa-LGAG first aggregates local features to obtain more representative statistical information before global feature extraction

$$\mathbf{X}'_{\text{spa}} = \text{LPA}\left(\text{Pad}_{\phi}\left(\mathbf{X}_{\text{spa}}\right)\right) \tag{6}$$

where $\mathbf{X}_{\text{spa}} \in \mathbb{R}^{C\times S\times S}$ represents the input feature; $\text{Pad}_{\phi}(\cdot)$ represents a boundary padding operation with a size of 1 in the spatial dimension. LPA(·) represents a 3×3 pooling operation with a sliding stride of 1, used to extract local statistical information. Parameter sensitivity experiments indicate that 11×11 is the optimal input patch size. Within this receptive field, the adoption of 3×3 local pooling aims to precisely capture substructural features; on the contrary, larger windows tend to cause local statistical information to converge toward regional averages, thereby losing fine textures. Subsequently, further global features are extracted, and dynamic enhancement of key features is achieved

$$\mathbf{X}^{\text{out}}_{\text{spa}} = \mathbf{X}_{\text{spa}} \otimes \text{Sig}\left(\text{GAP}_{\varphi}\left(\mathbf{X}'_{\text{spa}}\right)\right) \tag{7}$$

where $\text{GAP}_{\varphi}(\cdot)$ represents the average operation in the channel dimension, used to extract global statistical information. Sig(·) represents the sigmoid activation function used to generate weights and⊗ represents element wise multiplication.

2) Spe-LGAG: Similar to the design of Spa-LGAG, Spe-LGAG is proposed to enhance the expression of fine spectral features. But unlike the former, its local feature aggregation and feature weighting are both performed in the spectral dimension. The entire process can be expressed as follows:

$$\begin{aligned}\mathbf{X}'_{\text{spe}} &= \text{GAP}_{\phi}\left(\text{LPE}\left(\text{Pad}_{\varphi}\left(\mathbf{X}_{\text{spe}}\right)\right)\right)\\ \mathbf{X}^{\text{out}}_{\text{spe}} &= \mathbf{X}_{\text{spe}} \otimes \text{Sig}\left(\text{Linear}\left(\mathbf{X}'_{spe}\right)\right)\end{aligned} \tag{8}$$

where $\mathbf{X}_{\text{spe}} \in \text{R}^{C\times S\times S}$ represents the input feature, and $\text{Pad}_{\varphi}(\cdot)$ represents a boundary padding operation with a size of 1 in the spectral dimension. LPE(·) denotes a sliding average operation with a sliding step size of 1 and a window of adjacent three channels in the spectral dimension. $\text{GAP}_{\phi}(\cdot)$ stands for a global average pooling operation in the spatial dimension. linear(·) signifies a linear feature mapping layer.

The interaction between local and global features helps to improve the stability and robustness of attention weights, thereby more accurately highlighting key spatial spectral features.

D. Axial Recombination Spatial Mamba

Despite Mamba's outstanding performance in handling long sequence tasks, its original design intention is still focused on 1-D sequences. In HSIC, existing methods often use scanning strategies such as row-first, column-first, and cross scanning [48] to transform 2-D spatial features into 1-D sequences to adapt to Mamba. However, such transformations are prone to breaking the spatial topology structure, resulting in the disorderly splitting of feature units with strong semantic

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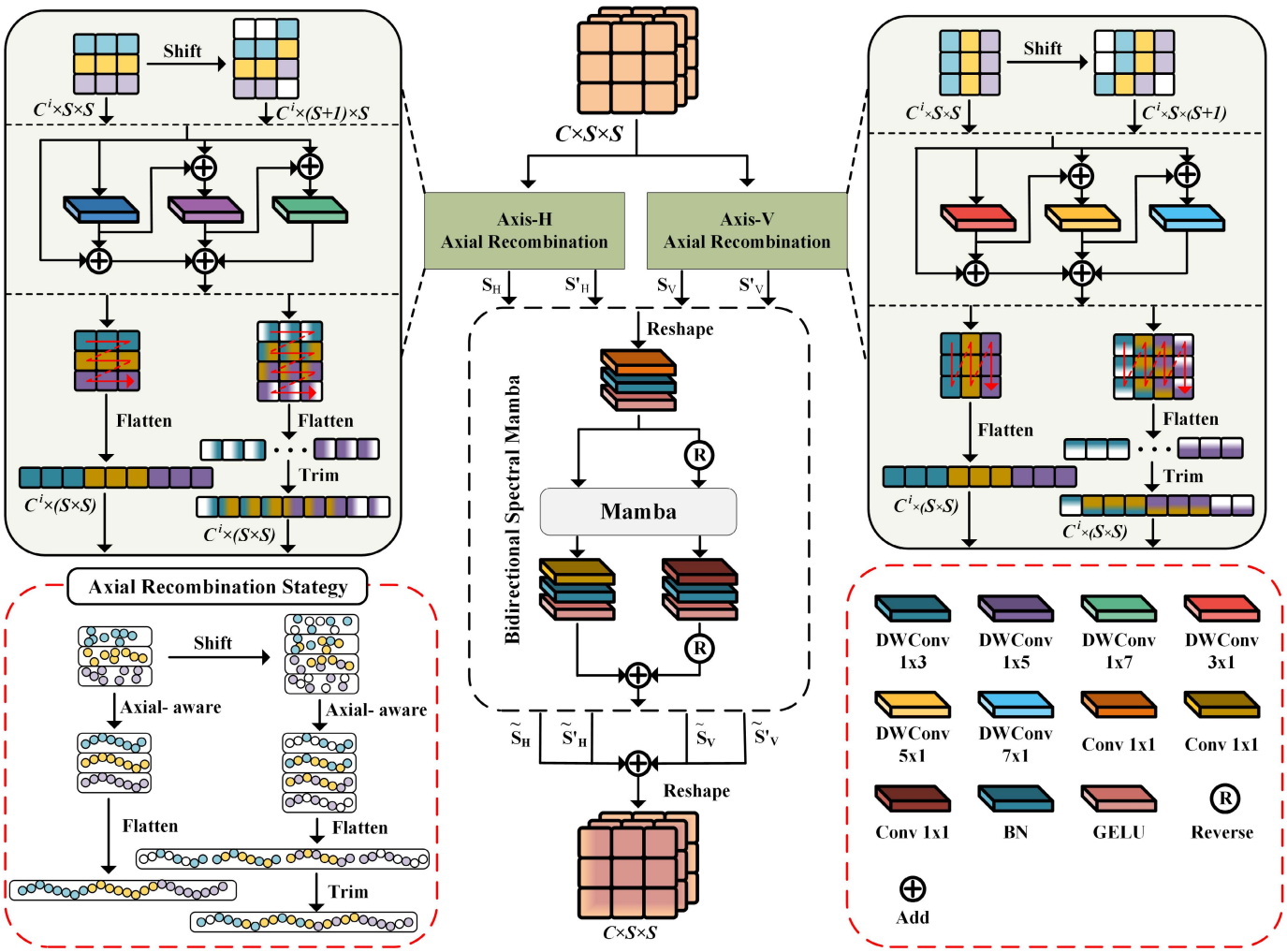


Fig. 4. Detailed structure of ARSM.

associations (such as adjacent pixels of the same object), making it difficult for Mamba to fully capture the global spatial contextual relationships between pixels. To this end, this article designs an ARSM module, whose structure is shown in Fig. 4. The module mainly consists of three parts: axial recombination, feature shift, and bidirectional spatial Mamba.

1) Axial Recombination: To enhance the semantic association in the generated sequence, axial awareness (based on horizontal and vertical axes) is first adopted to transform low-level pixel features into midlevel structural features. Taking the horizontal axis as an example, the process of axial perception is as follows:

$$\begin{aligned} \mathbf{Y}_1 &= \mathrm{DWConv}_1^{\rightarrow}(\mathbf{Y}) \\ \mathbf{Y}_{i+1} &= \mathrm{DWConv}_{i+1}^{\rightarrow}(\mathbf{Y}+\mathbf{Y}_i) \quad (i = 1,2) \end{aligned} \tag{9}$$

where $\mathbf{Y} \in \mathbb{R}^{C\times S\times S}$ represents the module input features. $\mathrm{DWConv}_k^{\rightarrow}(\cdot)$ represents strided convolution with a kernel size of $(2k+1)$, where $k = 1,2,3$, and $\rightarrow$ indicates along the horizontal direction. This design aims to model structured features under a multiscale receptive field. Subsequently, the feature map $\mathbf{Y}_{\mathrm{fuse}} = \mathbf{Y}+\mathbf{Y}_1+\mathbf{Y}_2+\mathbf{Y}_3$ is flattened along the horizontal direction to obtain a 1-D sequence $\mathbf{S}_{\mathrm{H}} \in \mathbb{R}^{C\times L}$, where L equals $S \times S$.

The axial recombination on the vertical axis is similar to the above process, using the vertical direction and independent training parameters to ultimately obtain a 1-D sequence $\mathbf{S}_{\mathrm{V}} \in \mathbb{R}^{C\times L}$.

The original spatial features belong to low-level representations, which directly encode the axial structural information of ground objects through axial recombination, providing more discriminative inputs for subsequent Mamba sequence modeling.

2) Feature Shift: Even with the introduction of the axial recombination strategy, it is still difficult to establish correlations between nonlocal features that cross row/column boundaries. Feature shift mainly refers to transferring boundary features from different rows/columns to the same row/column, so that subsequent axial perception can capture the boundary dependency relationships in the original feature map. In particular, taking the horizontal feature shift as an example (see the left side of Fig. 4), the feature in the ith row is denoted as $r_i \in \mathbb{R}^S$, where $0 \le i < S$. This process is

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represented as follows:

$$\begin{cases} r_i^{head}, r_i^{tail} = \mathrm{Div}\left(r_i, m\right) \\ r_i^{new}\left[m+1:\right] = r_i^{head} \\ r_{i+1}^{new}\left[:m+1\right] = r_i^{tail} \\ r_0^{new}\left[:m+1\right], r_{H+1}^{new}\left[m+1:\right] = 0 \end{cases} \tag{10}$$

where the partitioning operation Div(·) divides the row features into two parts, $r_i^{head} \in \mathbb{R}^m$ and $r_i^{tail} \in \mathbb{R}^n$, with parameter $m = S//2$ being the partitioning position and $n = S - m$. r^{new} represents a new row feature.

The initial feature map $\mathbf{Y}$ is transformed into feature map $\mathbf{Z}_{\mathrm{H}} \in \mathbb{R}^{C\times(S+1)\times S}$ through feature shifting. The feature representations in each row of $\mathbf{Z}_{\mathrm{H}}$ not only contain feature information that crosses row boundaries but also integrate contextual features from adjacent rows. By axially reorganizing $\mathbf{Z}_{\mathrm{H}}$, it helps to establish the correlation between cross-row-boundary features and adjacent row features. It should be noted that the 1-D sequence $\mathbf{S}_{\mathrm{H}} \in \mathbb{R}^{C\times(L+S)}$ obtained by axially sensing and flattening $\mathbf{Z}_{\mathrm{H}}$ needs to be cropped at both ends to match its dimension $\mathbb{R}^{C\times L}$.

For the vertical feature shift (as shown on the right side of Fig. 4), the process is like the horizontal feature shift, except that this operation shifts the column features to obtain the feature map $\mathbf{Z}_{\mathrm{V}} \in \mathbb{R}^{C\times S\times(S+1)}$.

3) Bidirectional Spatial Mamba: Mamba's unidirectional sequence processing approach may not fully capture potential bidirectional dependencies in HSIs. To this end, this study will invert the input sequence to obtain a reverse sequence and input the forward and reverse sequences separately into the parallel Mamba branch for feature modeling. Taking the 1-D sequence $\mathbf{S}_{\mathrm{H}}$ as an example

$$\begin{cases} \hat{\mathbf{S}}_{\mathrm{H}} = \mathit{Re}\mathrm{shape}(\mathrm{PWConv}_1(\mathit{Re}\mathrm{shape}(\mathbf{S}_{\mathrm{H}}))) \\ \hat{\mathbf{S}}_{\mathrm{rev}} = \mathit{Re}\mathrm{verse}\left(\mathrm{Permute}\left(\hat{\mathbf{S}}_{\mathrm{H}}\right)\right) \\ \mathbf{S}_{\mathrm{fuse}} = \mathrm{Permute}\big(\mathrm{PWConv}_2\left(\mathrm{Mamba}\left(\hat{\mathbf{S}}_{\mathrm{H}}\right)\right) \\ \qquad + \mathrm{PWConv}_3\left(\mathit{Mamba}\left(\hat{\mathbf{S}}_{\mathrm{rev}}\right)\right)\big) \end{cases} \tag{11}$$

where PWConv(·) represents the 2-D pointwise convolution operation. Reshape(·) represents shape reshaping operation. Permute(·) is used to change the order of data dimensions. Reverse(·) is used to generate a reverse input sequence. Mamba(·) represents the Mamba framework based on SSM, used to capture long-range spatial dependencies.

After axial recombination of feature maps $\mathbf{Z}_{\mathrm{H}}$ and $\mathbf{Z}_{\mathrm{V}}$, 1-D sequences $\mathbf{S}'_{\mathrm{H}} \in \mathbb{R}^{C\times L}$ and $\mathbf{S}'_{\mathrm{V}} \in \mathbb{R}^{C\times L}$ are obtained, respectively. Subsequently, they are passed through bidirectional space mamba along with sequences $\mathbf{S}_{\mathrm{H}}$ and S_{V} to obtain updated sequences $\tilde{\mathbf{S}}_{\mathrm{H}}, \tilde{\mathbf{S}}'_{\mathrm{H}}, \tilde{\mathbf{S}}_{\mathrm{V}}$, and $\tilde{\mathbf{S}}'_{\mathrm{V}}$. Finally, the above sequences are fused to generate the final output. The fusion process can be expressed as follows:

$$\mathbf{Y}_{\mathrm{out}} = \mathrm{Reshape}\left(\tilde{\mathbf{S}}_{\mathrm{H}} + \tilde{\mathbf{S}}'_{\mathrm{H}} + \tilde{\mathbf{S}}_{\mathrm{V}} + \tilde{\mathbf{S}}'_{\mathrm{V}}\right) \tag{12}$$

where reshape(·) represents reshaping a 1-D sequence into a 2-D feature map.

E. Group Offset Spectral Mamba

HSIs typically contain hundreds of continuous bands. Although band grouping can alleviate computational pressure, this process can disrupt the continuity of the original spectrum. This article designs a GOSM module, whose structure is shown in Fig. 5. The module mainly consists of two parts: group offset and bidirectional spectral Mamba.

1) Group offset: To avoid key information being fragmented during band grouping, this article introduces a group offset mechanism, which allows key information that crosses group boundaries to be fully captured in other offset views, thereby alleviating the problem of spectral continuity being disrupted. For the input spectral sequence $\mathbf{T} \in \mathbb{R}^{C_4}$, the process can be represented as follows:

$$\begin{cases} \mathbf{G}_1 = \{g_1, g_2, \ldots, g_n\} = \mathrm{Div}\left(\mathbf{T}, l\right) \\ \mathbf{G}_{\mathrm{i}} = \{g_{i_1}, g_{i_2}, \ldots, g_{i_n}\} = \mathrm{Offset}\left(\mathbf{G}_1, k_i\right) \quad (i = 2, 3, 4) \end{cases} \tag{13}$$

where Div(·) represents dividing the spectral sequence into multiple groups and l represents the length of each group. g_i is the ith spectral subgroup, $\mathrm{G}_{\mathrm{i}} \in \mathbb{R}^{n\times l}$ represents the ith spectral grouping sequence, and n is equal to $C//l$. Offset(·) is used to offset the grouped sequence. The offset k is taken as $\{0, 1/4l, 2/4l, 3/4l\}$ and by constructing four sets of offset sequences, this design achieves the repositioning of edge bands within the groups, ensuring that the continuity information across boundaries is fully modeled in the multiview representation. This scheme maintains a balance between enhancing the representational power of the model and controlling computational overhead. Furthermore, the design of dynamically scaling k with the group length l ensures that the scheme can naturally adapt to various datasets with different spectral dimensions.

2) Bidirectional Spectral Mamba: To capture potential bidirectional dependencies in spectral sequences, the input sequence is first inverted to generate a reverse sequence and then the forward and reverse sequences are modeled separately. Taking the grouping sequence $\mathbf{G}_1$ as an example, this process can be formulated as follows:

$$\mathbf{G}'_1 = \mathit{PWConv}_4\left(\mathbf{G}_1\right), \quad \mathbf{G}'_{1,\mathit{rev}} = \mathit{Reverse}\left(\mathbf{G}'_1\right) \tag{14}$$

$$\mathbf{G}_{\mathit{fuse}} = \mathit{PWConv}_5\left(\mathit{Mamba}\left(\mathbf{G}'_1\right)\right) + \mathit{PWConv}_6\left(\mathit{Mamba}\left(\mathbf{G}'_{1,\mathit{rev}}\right)\right) \tag{15}$$

where $\mathit{PWConv}(\cdot)$ denotes a 1-D pointwise convolution operation, and different pointwise convolutional layers do not share parameters. It should be noted that this convolutional layer does not introduce an activation function. The design of linear convolution circumvents the nonlinear truncation effect by preserving spectral continuity and homogeneous mapping characteristics, thereby enhancing the model's robustness to illumination fluctuations. This approach aligns with the pixel linear mixing model, effectively preventing physical semantic distortion and ensuring the authenticity of proportional relationships while focusing on spectral morphological features. $\mathit{Reverse}(\cdot)$ represents reversing the grouping sequence. $\mathit{Mamba}(\cdot)$ represents the Mamba framework.

After passing through the bidirectional spectral Mamba, the four spectral sequences are modeled into grouped sequences $\tilde{\mathbf{G}}_1, \tilde{\mathbf{G}}_2, \tilde{\mathbf{G}}_3$ and $\tilde{\mathbf{G}}_4$. Before sequence fusion, the grouped sequences need to be reverse-shifted to ensure accurate

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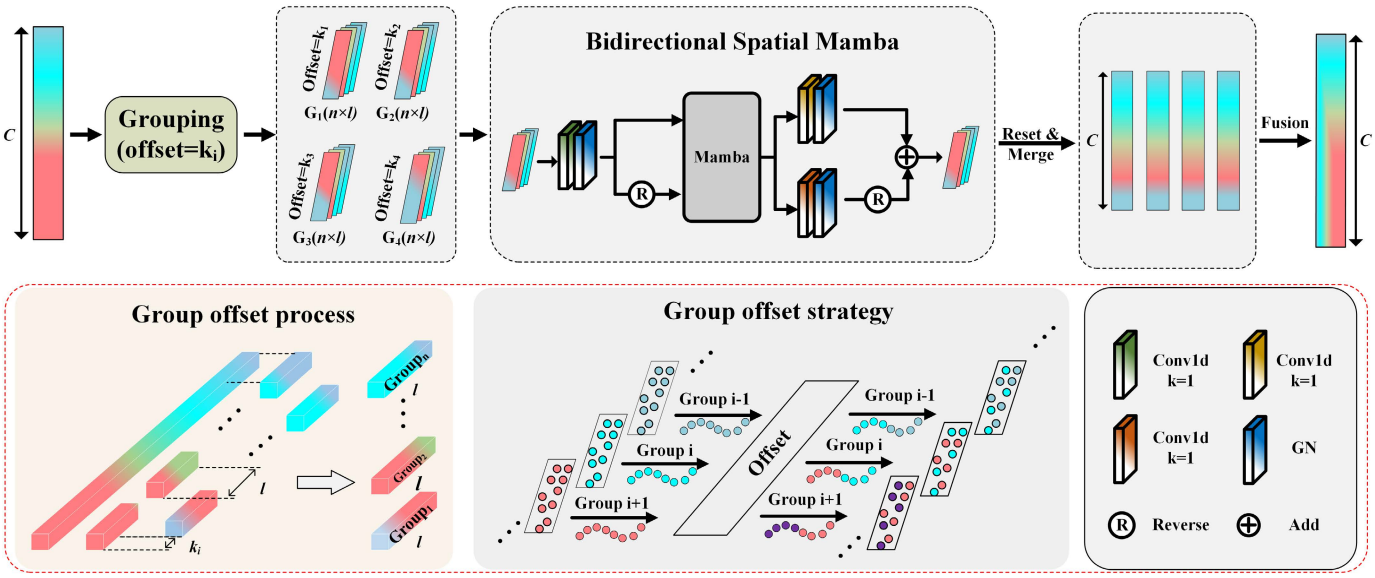


Fig. 5. Detailed structure of GOSM.

TABLE I
DETAILS OF THE EXPERIMENTAL DATASETS

Dataset	IP	PU	HST	LK
Size (m)	145x145	610×340	340×1905	550×400
Spatial resolution	20	1.3	2.5	0.463
Number of bands	200	103	144	270
Number of classes	16	9	15	9
Training proportion (%)	3	0.6	3	0.15

alignment in the spectral dimension during fusion. This process is represented as follows:

$$\begin{cases} \tilde{\mathbf{G}}_i' = \text{Reset}\left(\tilde{\mathbf{G}}_i, k_i\right), \\ \mathbf{T}_i = \text{Merge}\left(\tilde{\mathbf{G}}_i'\right), \end{cases} \quad (i = 1, 2, 3, 4) \tag{16}$$

where Reset(·) represents the reverse group offset of the grouping sequence and k_i represents the offset, which corresponds to the value in (13). Merge(·) represents merging the sequence of groups, with each group being concatenated one by one in sequence. Finally, the fused sequence $\mathbf{T}_{\text{out}} = \mathbf{T}_1 + \mathbf{T}_2 + \mathbf{T}_3 + \mathbf{T}_4$ is obtained. The fused feature $\mathbf{T}_{\text{out}} \in \mathbb{R}^C$ integrates complementary information from different spectral perspectives, thereby forming a more comprehensive and robust spectral representation.

IV. Experimental Results and Analysis

This section will first introduce the dataset used in the experiment, the hardware configuration of the experiment, and the parameter settings. Afterward, the proposed method will be comprehensively validated through quantitative evaluation, visual evaluation, and ablation experiments.

A. Experimental Setup

1) Datasets: This article selects four benchmark hyperspectral datasets covering different application scenarios: Indian Pines (IP) [51] and WHU-Hi-Longkou (LK) [53], which contain agricultural land features, as well as Pavia University (PU) [51] and Houston 2013 (HST) [52], which represent typical urban landscapes. The basic information of each dataset (including spatial resolution, spectral range, and number of classes) is shown in Table I. For all datasets, training samples are selected using a stratified random sampling strategy: for each class, a fixed proportion of labeled pixels is randomly drawn for training, with a minimum of 3 pixels per class to guarantee that every class, including the smallest ones, is represented under low training ratios.

2) Implementation Details: In the PyCharm development environment, the proposed method is implemented based on Python 3.10.14 and PyTorch 2.1.2. All experiments in this article are conducted under the PyTorch framework and run on a server equipped with a 2.1 GHz Intel[1] Xeon[1] Silver 8352V CPU and a 24 GB Nvidia GeForce RTX 4090 GPU.

To objectively evaluate model performance, overall accuracy (OA), average accuracy (AA), and kappa coefficient (K) were used as evaluation metrics. In the proposed method, the Adam optimizer was employed, with the number of training epochs set to 300. In GOSM, the group size was set to 14, and the offset was set to 4 each time. The hidden state dimension in the Mamba architecture was set to 16. The training set was randomly sampled according to the proportions listed in Table I, and the remaining pixels were used for testing. The detailed division of the training and testing sets is presented in Table II.

B. Performance Comparison

To further validate the effectiveness of the proposed method, this article selected three representative HSIC architectures for comprehensive comparison, including CNN based methods (HybridSN [34], PyResNet [32]), Transformer based methods (HIT [35], SSTN [41], SSFTT [36], A^{2}S^{2}KResNet [37]), and

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TABLE II

CATEGORY NAMES AND SAMPLE SIZES OF THE TRAINING AND TESTING SETS FOR EACH CATEGORY ON THE IPS, PU, HST, AND LK DATASETS

	Indian Pines			Pavia University			Houston 2013			WHU-Hi-LongKou		
No.	Category	Train	Test	Category	Train	Test	Category	Train	Test	Category	Train	Test
1	Alfalfa	3	43	Asphalt	40	6591	Healthy grass	37	1214	Corn	52	34459
2	Corn-notill	43	1385	Meadows	112	18537	Stressed grass	38	1216	Cotton	12	8362
3	Corn-mintill	25	805	Gravel	12	2087	Synthetic grass	21	676	Sesame	4	3027
4	Corn	7	230	Trees	18	3046	Trees	37	1207	Broad-Leaf Soybean	95	63117
5	Grass-pasture	14	469	Painted metal sheets	8	1337	Soil	37	1205	Narrow-Leaf Soybean	6	4145
6	Grass-trees	22	708	Bare soil	30	4999	Water	10	315	Rice	18	11836
7	Grass-pasture-mowed	3	25	Bitumen	8	1322	Residential	38	1230	Water	100	66956
8	Hay-windrowed	14	464	Self-blocking bricks	22	3660	Commercial	37	1207	Roads and Houses	11	7113
9	Oats	3	17	Shadows	6	941	Road	37	1215	Mixed Weed	8	5221
10	Soybean-notill	29	943				Highway	37	1190			
11	Soybean-mintill	73	2382				Railway	37	1198			
12	Soybean-clean	18	575				Parking Lot 1	37	1196			
13	Wheat	6	199				Parking Lot 2	14	455			
14	Woods	38	1227				Tennis Court	13	415			
15	Buildings	12	374				Running Track	20	640			
16	Stone	3	90									
	Total	313	9936	Total	256	42520	Total	450	14579	Total	306	204236

.

TABLE III

QUANTITATIVE RESULT (ACC% ± STD%) OF IP. BEST IN BOLD

Class	CNN-based Methods		Transformer-base Methods				Mamba-base Methods				proposed
	HybridSN	PyResNet	HIT	SSTN	SSFTT	A^2S^2KResNet	3DSSMamba	MambaLG	MambaHSI	Local-VMamba	
OA(%)	78.68±2.36	71.53±4.61	70.31±2.20	92.85±1.52	93.79±0.6	94.82±1.26	77.14±2.31	91.93±1.33	95.54±0.82	93.35±1.20	**96.94±0.49**
AA(%)	70.74±5.96	75.75±2.91	71.08±3.40	91.67±3.20	90.26±1.61	90.62±0.99	67.89±2.65	75.57±5.04	94.05±0.87	92.88±1.35	**96.28±0.93**
K(%)	75.55±2.77	67.05±5.56	65.66±2.55	91.85±1.73	92.90±0.69	94.10±1.43	73.94±2.67	90.77±1.53	94.5±1.85	92.41±1.38	**96.51±0.56**
1	41.97±28.56	90.16±7.80	67.26±15.59	**100±0**	43.17±30.77	87.05±9.59	20.32±16.52	28.57±22.16	90.82±5.89	85.05±14.86	95.35±5.27
2	72.94±6.34	56.24±10.84	64.67±3.11	90.10±5.82	85.96±1.25	94.03±1.11	76.51±2.57	94.51±1.30	93.26±1.67	89.99±3.47	**96.27±1.99**
3	81.41±5.45	69.89±11.95	62.74±7.27	91.06±2.93	90.70±3.03	94.64±3.05	66.69±5.49	87.15±1.40	90.48±1.63	90.98±6.77	**96.15±1.66**
4	63.80±7.25	62.35±14.75	63.05±14.71	91.71±8.88	91.24±2.35	91.46±4.38	55.65±10.88	78.32±7.32	85.65±9.98	89.50±5.49	**92.98±2.93**
5	90.27±3.65	91.20±6.88	77.22±8.41	98.99±0.92	**99.60±0.35**	98.04±1.60	81.30±14.00	90.43±0.97	89.80±2.69	87.60±3.43	90.31±3.10
6	95.57±2.14	88.48±7.27	77.87±6.48	96.90±1.91	**99.19±0.38**	97.89±1.16	79.20±7.50	94.04±2.58	97.55±1.35	98.95±0.65	98.73±0.43
7	51.38±27.19	71.66±19.76	80.43±10.17	82.22±19.66	**100.00**	72.72±22.10	56.61±5.50	18.68±22.75	99.4±1.46	97.14±4.64	98.29±4.20
8	94.17±4.57	93.99±3.28	96.27±1.72	97.81±3.11	99.66±0.53	99.81±0.30	90.89±3.81	99.47±0.70	98.32±1.9	97.11±5.70	**99.97±0.07**
9	31.68±15.08	44.01±28.59	57.02±21.94	62.99±43.08	89.47±6.29	56.13±15.14	47.47±16.40	0.79±1.94	**100±0.00**	98.32±2.66	98.32±4.12
10	67.89±7.72	73.49±15.23	60.85±7.66	88.22±4.72	92.18±1.18	90.59±6.10	77.31±4.14	92.33±1.41	92.12±2.69	90.59±2.09	**94.73±1.69**
11	77.31±3.40	69.30±8.77	68.00±2.43	95.72±4.14	97.33±0.51	95.91±1.75	85.14±2.02	95.39±1.20	97.11±2.46	95.24±2.69	**98.46±0.73**
12	65.80±7.46	52.35±11.77	54.23±5.90	87.89±6.71	83.83±3.77	93.70±4.08	63.45±3.29	79.82±5.72	86.46±4.35	88.84±4.67	**95.38±2.56**
13	85.35±20.24	91.29±8.59	85.16±4.88	98.35±2.01	**100.00**	97.88±1.41	74.66±14.78	98.11±1.04	98.00±1.72	97.49±4.59	96.98±3.46
14	91.25±3.25	87.46±6.46	82.78±2.59	93.31±3.94	98.87±0.62	97.21±2.37	82.85±8.47	98.22±0.39	98.5±0.9	98.79±0.74	**99.62±0.52**
15	75.31±5.07	82.91±6.41	69.16±5.76	92.56±2.45	92.09±4.72	91.14±5.95	51.15±7.34	83.11±7.56	**97.53±2.54**	87.55±6.11	95.07±3.43
16	45.70±29.21	87.27±9.61	70.61±10.20	**98.93±1.42**	80.79±8.31	91.73±5.01	77.14±19.47	70.14±37.23	89.77±6.25	92.86±6.72	93.81±9.59

TABLE IV

QUANTITATIVE RESULT (ACC% ± STD%) OF PU. BEST IN BOLD

Class	CNN-based Methods		Transformer-base Methods				Mamba-base Methods				proposed
	HybridSN	PyResNet	HIT	SSTN	SSFTT	A^2S^2KResNet	3DSSMamba	MambaLG	MambaHSI	Local-Vmamba	
OA(%)	81.14±4.14	87.27±1.99	81.21±1.88	95.36±0.94	95.49±0.48	95.74±0.69	87.39±1.68	95.66±0.91	96.09±0.47	96.81±0.85	**98.59±0.27**
AA(%)	73.54±6.56	85.80±1.62	75.36±1.96	95.25±1.40	92.44±1.14	94.88±0.80	78.41±2.90	90.20±3.71	93.03±0.85	94.88±1.40	**97.15±0.71**
K(%)	74.92±5.45	82.87±2.70	74.66±2.60	93.82±1.30	94.00±0.64	94.34±0.93	83.27±2.23	94.20±1.25	94.86±0.97	95.76±1.13	**98.13±0.36**
1	77.34±6.86	82.48±6.99	76.47±4.13	93.72±1.80	94.67±1.06	93.02±4.21	88.45±3.16	98.79±0.61	97.11±1.29	95.37±2.37	**99.41±0.48**
2	93.20±1.76	91.97±1.53	88.99±1.88	97.67±1.16	99.91±0.02	98.72±0.48	96.76±1.18	99.76±0.10	99.65±0.25	99.42±0.21	**99.97±0.05**
3	64.32±11.80	63.55±8.61	44.09±4.54	84.27±15.60	77.25±2.61	86.63±9.17	57.68±9.81	88.76±3.19	87.10±6.44	86.84±8.35	**92.88±3.43**
4	70.32±16.95	96.29±1.98	92.17±3.04	96.99±1.52	90.19±1.93	**98.76±1.33**	79.29±7.02	94.10±1.42	79.04±7.76	94.20±1.79	94.29±1.73
5	89.08±7.74	95.53±3.52	93.74±3.54	98.97±0.70	99.40±0.48	99.00±0.83	98.39±0.61	97.76±1.36	**99.65±0.55**	99.47±0.87	99.47±0.90
6	80.95±6.61	90.09±5.93	73.32±7.74	98.93±0.97	96.92±1.30	98.46±1.27	82.53±7.52	94.69±0.76	99.18±0.73	97.47±2.17	**99.69±0.10**
7	65.03±23.46	81.47±7.66	49.03±6.85	99.52±0.60	**99.97±0.08**	96.69±2.95	62.11±7.53	77.06±5.64	87.08±7.11	92.30±3.35	99.60±0.37
8	63.79±5.55	77.84±5.11	70.89±4.15	88.41±5.55	86.49±2.34	84.11±4.10	80.55±3.94	87.51±3.82	93.6±3.22	94.34±1.68	**96.35±2.06**
9	57.82±20.10	92.98±6.06	89.50±5.85	**98.75±0.91**	87.17±5.70	98.50±1.82	59.97±14.01	73.36±31.46	94.87±2.85	94.52±3.20	92.67±5.46

Mamba based methods (3DSSMamba [43], MambaLG [46], MambaHSI [31], Local-VMamba [54]). As Local-VMamba is not inherently tailored for HSI classification, we adapted it based on the MambaHSI framework. Tests indicate that a stack of two basic modules yields optimal performance. To ensure the fairness of the results, all experiments were conducted under the same conditions, and the final result was taken as the average of seven consecutive independent experiments.

1) Quantitative Experimental Results and Analysis: Tables III–VI show the classification results of each method on four datasets: IPs, PU, HST, and WHU Hi Longkou, including OA, AA, K, and precision for each category. It can be seen that the proposed ARGO-Mamba is significantly better than

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TABLE V
QUANTITATIVE RESULT (ACC% ± STD%) OF HST. BEST IN BOLD

Class	CNN-based Methods		Transformer-base Methods				Mamba-base Methods				proposed
	HybridSN	PyResNet	HIT	SSTN	SSFTT	A^2S^2KResNet	3DSSMamba	MambaLG	MambaHSI	Local-Vmamba	
OA(%)	89.43±1.09	92.17±1.74	83.49±1.24	89.53±2.09	95.25±0.98	96.37±0.70	91.27±1.63	95.38±0.79	94.72±0.48	95.72±0.58	**96.93±0.32**
AA(%)	90.78±1.07	93.35±1.08	83.82±1.95	91.27±1.46	95.50±1.11	96.77±0.70	91.80±1.51	95.03±0.97	94.66±0.59	95.90±0.67	**96.88±0.42**
K(%)	88.58+1.18	91.54+1.89	82.12±1.34	88.68±2.26	94.86±1.06	96.08±0.76	90.56±1.76	95.00±0.85	94.11±0.93	95.38±0.63	**96.69±0.35**
1	92.47±2.17	91.21±3.43	86.58±3.72	88.65±4.77	96.94±1.47	96.28±2.38	96.16±1.51	98.02±3.49	96.36±3.85	**98.35±1.26**	98.21±0.72
2	95.06±4.20	95.50±1.88	90.55±2.39	97.75±2.16	97.70±1.59	**97.94±1.82**	92.72±3.20	96.36±1.06	97.76±1.17	97.10±1.89	**97.94±1.73**
3	98.96±1.51	98.63±2.08	92.38±2.54	96.28±5.66	99.30±1.18	99.81±0.31	98.92±0.49	99.77±0.28	99.72±0.52	**99.83±0.12**	99.56±0.30
4	93.92±5.50	94.67±2.12	89.19±3.69	95.64±6.73	98.27±1.36	97.56±1.45	91.74±3.15	**99.21±0.52**	97.62±1.98	95.63±2.05	98.28±1.99
5	96.28±3.78	97.61±2.34	92.38±3.03	98.97±1.47	99.83±0.26	98.86±1.68	97.75±1.19	99.80±0.29	**100.00**	99.83±0.41	**100.00**
6	96.99±5.02	95.87±4.11	80.70±9.99	95.67±8.76	96.28±5.30	**100.00**	88.80±2.94	84.62±5.33	87.2±8.39	89.80±5.19	94.78±3.92
7	84.68+5.50	95.80±1.79	84.91±2.64	90.19±4.99	89.61±7.09	95.02±2.65	84.89±2.96	95.85±1.45	94.61±2.21	94.05±3.12	**96.00±1.81**
8	90.65±7.51	95.06±2.79	83.65±6.10	94.08±3.28	91.18±4.67	**98.25±2.64**	81.15±2.93	86.76±2.55	84.19±5.13	90.35±3.00	93.94±2.01
9	87.83±5.41	86.50±4.64	77.85±4.25	88.35±4.80	89.39±4.64	**93.98±4.09**	83.60±5.49	91.83±1.03	92.17±2.18	90.18±4.23	91.50±4.14
10	82.05±3.00	81.93±6.95	77.75±2.29	75.40±7.05	92.97±6.64	92.00±4.01	92.59±3.24	98.13±1.36	96.73±3.01	96.22±1.91	**99.33±0.97**
11	94.14±9.20	92.39±1.82	78.36±4.06	87.23±13.44	**98.15±2.81**	96.60±1.77	91.30±4.27	94.85±2.50	92.84±2.09	96.82±1.91	97.97±1.58
12	91.92±6.27	87.48±9.10	73.08±2.98	82.80±9.45	94.41±2.83	94.76±3.03	92.16±3.31	92.10±3.54	92.7±3.76	94.05±2.97	**95.00±3.58**
13	86.85+8.54	**96.34±3.12**	80.15±7.34	84.35±8.53	89.29±4.89	92.86±4.53	87.25±5.00	90.26±3.90	93.2±2.29	96.26±2.24	90.68±5.19
14	93.03±10.23	94.46±3.25	84.76±6.67	98.27±2.71	99.31±1.16	98.20±2.28	99.04±0.66	98.69±2.33	99.72±0.6	**100.00**	**100.00**
15	95.82±2.30	96.87±1.82	85.01±3.85	95.37±2.92	99.89±0.16	99.46±0.67	98.88±1.66	99.20±1.41	95.08±4.00	**100.00**	**100.00**

TABLE VI
QUANTITATIVE RESULT (ACC% ± STD%) OF LK. BEST IN BOLD

Class	CNN-based Methods		Transformer-base Methods				Mamba-base Methods				proposed
	HybridSN	PyResNet	HIT	SSTN	SSFTT	A^2S^2KResNet	3DSSMamba	MambaLG	MambaHSI	Local-Vmamba	
OA(%)	86.92±5.34	96.27±0.70	92.41±0.52	96.86±1.10	98.21±0.17	97.93±0.44	92.78±1.82	94.53±2.24	97.28±0.40	97.42±0.57	**98.96±0.12**
AA(%)	73.70±16.87	94.95±1.14	80.64±2.30	93.38±3.21	93.56±0.79	95.64±1.73	77.10±6.13	77.06±8.84	90.22±2.70	91.05±2.35	**95.90±0.21**
K(%)	82.73±7.07	95.06±0.93	89.98±0.69	95.85±1.47	97.64±0.23	97.28±0.57	90.46±2.43	92.77±2.96	93.21±3.86	96.61±0.74	**98.63±0.16**
1	79.24±10.27	97.40±2.27	98.34±1.29	99.48±0.28	99.82±0.13	99.24±0.58	96.89±1.24	99.19±0.62	99.55±0.26	99.89±0.13	**99.95±0.04**
2	64.10±22.25	93.12±3.71	65.27±6.99	85.62±11.89	97.20±0.86	94.72±2.87	83.81±9.96	80.90±33.12	94.17±3.45	94.58±3.27	**98.21±1.29**
3	55.11±47.85	95.72±4.64	77.69±15.20	99.62±0.78	97.48±1.02	**99.99±0.03**	46.09±29.80	44.69±21.96	78.16±5.82	88.96±4.67	87.97±3.89
4	90.61±6.41	94.08±2.52	92.06±2.03	96.32±2.36	**99.71±0.09**	97.93±0.87	95.77±1.98	98.46±1.06	98.28±0.47	98.58±0.55	99.47±0.29
5	54.22±39.92	93.25±3.86	38.17±6.15	82.49±21.72	74.81±5.28	**99.10±0.67**	49.47±16.96	32.71±37.17	72.29±17.38	68.11±21.95	92.85±1.77
6	87.80±11.59	98.25±1.09	96.62±2.50	99.50±0.74	98.43±0.40	99.10±0.61	83.95±12.45	98.70±0.93	98.29±1.61	98.09±1.72	**99.53±0.66**
7	98.45±2.20	98.90±1.04	99.07±0.56	99.37±0.41	99.54±0.33	99.56±0.23	99.45±0.28	99.89±0.08	99.93±0.09	99.71±0.31	**99.96±0.04**
8	71.45±18.74	91.77±4.20	78.80±5.13	85.06±6.14	85.73±3.89	84.84±4.24	76.64±7.05	76.02±17.29	85.43±3.83	83.17±7.06	**94.43±0.87**
9	62.31±39.06	92.10±4.98	79.72±7.02	**92.99±5.17**	89.34±4.04	86.25±10.09	61.85±13.15	62.94±10.30	85.91±9.22	88.38±4.93	90.75±3.48

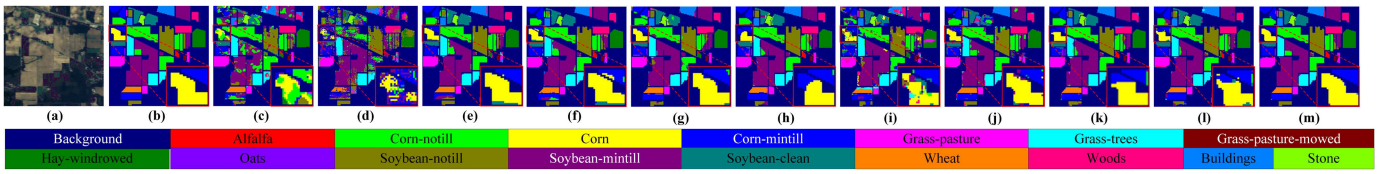


Fig. 6. Classificationmaps obtained by different methods on the IP dataset. (a) Pseudocolor image, (b) ground truth map, (c)–(l) HybridSN, PyResNet, HIT, SSTN, SSFTT, A2S2KResNet, 3DSSMamba, MambaLG, MambaHSI, and Local-VMamba, respectively, and (m) proposed.

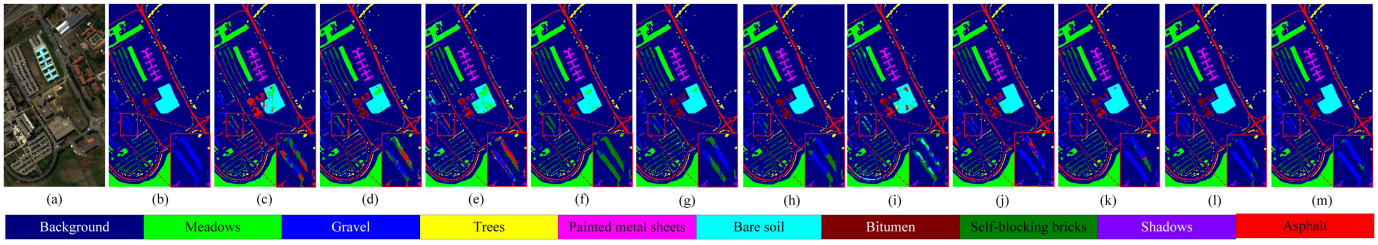


Fig. 7. Classification maps obtained by different methods on the PU dataset. (a) Pseudocolor image, (b) ground truth map, (c)–(l) HybridSN, PyResNet, HIT, SSTN, SSFTT, A^2S^2KResNet, 3DSSMamba, MambaLG, MambaHSI, and Local-VMamba, respectively, and (m) proposed.

other methods in all three main indicators, and the results have the smallest fluctuations, with over half of the categories achieving optimal accuracy. Overall, CNN-based methods are limited by local receptive fields and struggle to capture global contextual information, resulting in the worst performance. For example, for the IPs dataset, the OA of the optimal CNN method is 18.26% lower than that of ARGO-Mamba. On the contrary, methods based on Transformer and Mamba have achieved significant advantages, which strongly demonstrates the criticality of global context modeling for HSIC. For the PU dataset, ARGO-Mamba continued its previous advantages and achieved the best classification results in five categories. This indicates that the proposed axial restructuring strategy can more effectively mine spatial information for urban scenes with complex results. For HST, compared to the best of the three methods, ARGO-Mamba's OA values increased

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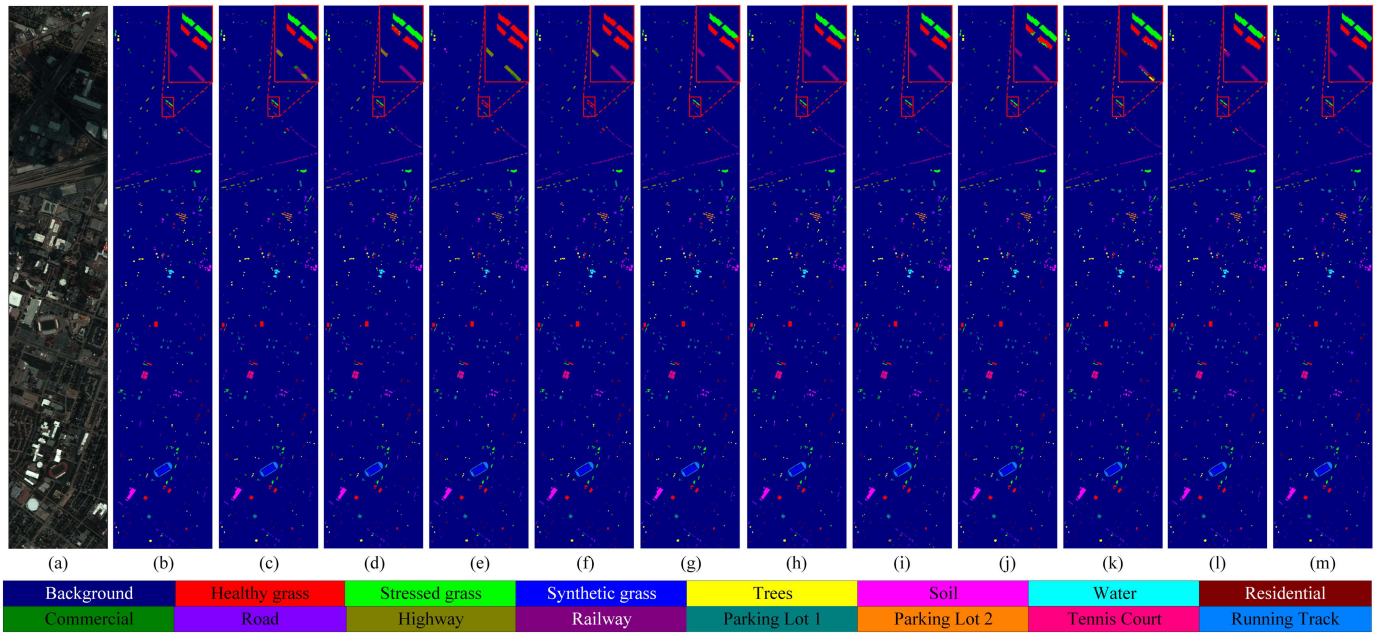


Fig. 8. Classification maps obtained by different methods on the HST dataset. (a) Pseudocolor map, (b) ground truth map, (c)–(l) HybridSN, PyResNet, HIT, SSTN, SSFTT, A^{2}S^{2}KResNet, 3DSSMamba, MambaLG, MambaHSI, Local-VMamba, respectively, and (m) proposed.

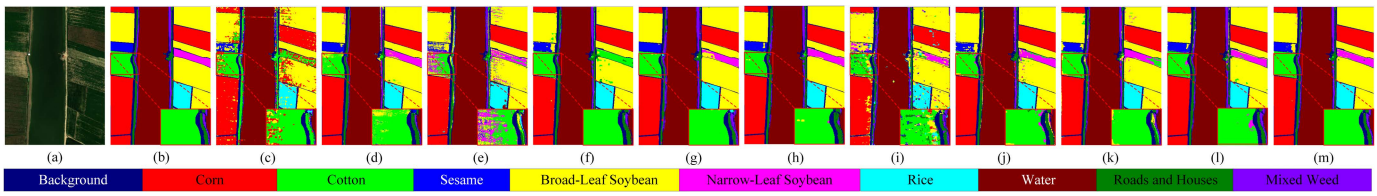


Fig. 9. Classification maps obtained by different methods on the LK dataset. (a) Pseudocolor map, (b) ground truth map, (c)–(l) are, respectively, HybridSN, PyResNet, HIT, SSTN, SSFTT, A2S2KResNet, 3DSSMamba, MambaLG, MambaHSI, Local-VMamba, and (m) proposed.

TABLE VII
IMPACT OF DIFFERENT COMPONENTS ON OA. BEST IN BOLD

Case	Component					Dataset			
	Baseline	GOSM	SPE-LGAG	ARSM	SPA-LGAG	IP	PU	HST	LK
1	✓	×	×	×	×	92.12	94.85	93.86	97.69
2	✓	✓	×	×	×	95.23	96.62	95.28	98.19
3	✓	✓	✓	×	×	95.34	96.71	95.56	98.27
4	✓	×	×	✓	×	96.77	98.26	96.18	98.46
5	✓	×	×	✓	✓	96.83	98.41	96.54	98.58
6	✓	✓	✓	✓	✓	**96.94**	**98.59**	**96.93**	**98.96**

by 4.76%, 0.56%, and 1.55%, respectively. Among them, category 5 (land), category 14 (tennis court), and category 15 (runway) achieved error-free classification, which is also attributed to the processing of spatial and spectral features by axial recombination and group offset strategies. For LK, compared to the best of the three methods, ARGO-Mamba's OA values increased by 2.69%, 0.75%, and 1.68%, respectively, and achieved the best results in the five categories. The above results fully verify the robustness and generalization ability of the proposed method in different scenarios.

2) Visualization Results and Analysis: Figs. 6–9 show the visualization classification results of different methods on four datasets. It can be seen that ARGO-Mamba generated classification maps that are highly consistent with the original images on all datasets, with clear overall structure and high fidelity to local details (such as the corn area in IPs, the weed area in PU, the healthy and pressure grass area in HST, and the cotton area in LK).

On the contrary, CNN-based methods perform significantly less well, with blurry edges and more noise in their classification maps. The local magnification results further indicate that ARGO-Mamba exhibits higher accuracy and consistency in the characterization of land features, effectively distinguishing similar land features. This advantage stems from the model's early use of attention mechanisms to focus on key features, thereby suppressing noise interference. At the same time, the axial reorganization strategy can deeply explore the semantic associations of land features and enhance the perception of complex land feature relationships. The group offset strategy further enhances the ability to model local features, which helps distinguish vegetation and targets with similar spectra.

4) t-SNE Resultsand Analysis: To visually analyze the feature representations learned by the model, this article uses t-distributed stochastic neighbor embedding (t-SNE) [50] to project high-dimensional features into a 2-D space. Fig. 10 shows the visualization results on the different datasets and compares them with CNN-based PyResNet, Transformer-based A2S2KResNet, and Mamba-based MambaHSI. It can

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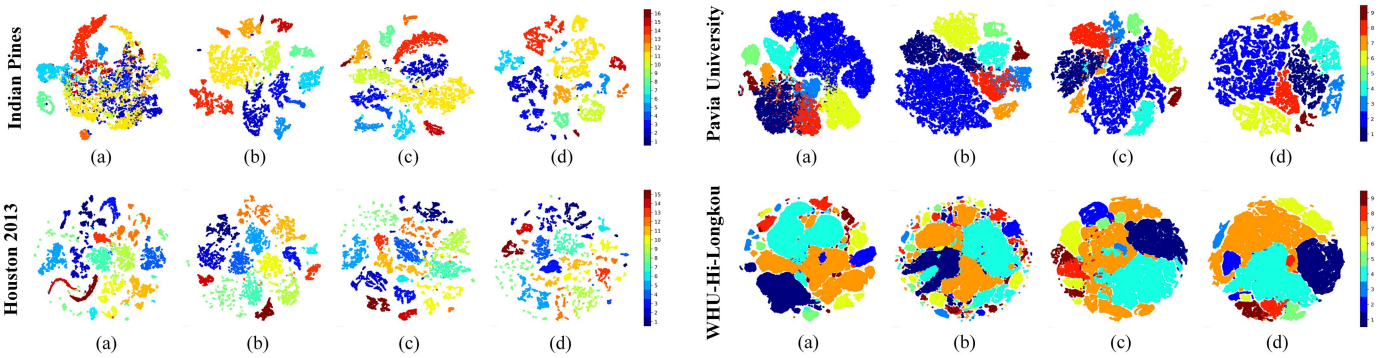


Fig. 10. t-SNE visualization on the IPs, PU, HST, and LK datasets. For each dataset, (a)–(d) correspond to PyResNet, A^{2}S^{2}KResNet, MambaHSI, and the proposed method, respectively.

TABLE VIII
COMPUTATIONAL COMPLEXITY ANALYSIS OF DIFFERENT METHODS

		HybridSN	PyResNet	HIT	SSTN	SSFTT	A^{2}S^{2}KResNet	3DSSMamba	MambaLG	MambaHSI	Local-VMamba	Proposed
IP	Params(M)	0.51	21.86	56.81	0.02	0.15	0.37	0.02	0.19	0.43	0.35	0.4
	Flops(G)	0.55	2.12	24.23	0.32	0.76	2.75	1.71	3.59	4.66	2.56	2.14
	Testing time(s)	0.25	1.65	1.92	0.41	0.14	1.57	0.60	0.05	0.01	0.05	0.85
PU	Params(M)	1.89	21.85	48.22	0.02	0.15	0.22	0.02	0.19	0.41	0.34	0.4
	Flops(G)	2.46	2.12	13.54	0.29	0.76	1.41	1.71	32.83	43.51	2.51	2.09
	Testing time(s)	1.68	7.12	8.08	1.53	0.58	4.52	2.50	0.15	0.12	0.01	3.19
HST	Params(M)	2.65	21.85	51.18	0.02	0.15	0.28	0.02	0.19	0.42	0.34	0.4
	Flops(G)	3.56	2.12	17.18	0.3	0.76	1.97	1.71	107.41	142.92	2.53	2.12
	Testing time(s)	0.68	2.40	2.71	0.51	0.21	1.89	0.87	0.46	0.43	0.04	1.15
LK	Params(M)	4.97	21.85	66.64	0.02	0.15	0.48	0.02	0.2	0.43	0.35	0.41
	Flops(G)	6.92	2.12	36.6	0.35	0.76	3.73	1.71	37.14	50.86	2.59	2.18
	Testing time(s)	52.37	34.713	41.45	38.13	2.86	47.23	12.04	0.21	0.40	0.04	20.05

be seen that ARGO-Mamba exhibits excellent clustering performance on all datasets. In particular, PyResNet has the most severe feature distribution confusion. Although A2S2KResNet and MambaHSI can form certain class clusters, there is still some class overlap. On the contrary, ARGO-Mamba can tightly cluster similar samples and form clear boundaries between different categories, indicating that its learned features have higher intraclass consistency and interclass separability. This result further confirms the significant advantage of the model in spatial–spectral joint feature discrimination ability.

C. Ablation Study

The experimental results are shown in Table VII. From the table, it can be seen that the model with only baseline performs the worst on all datasets. When GOSM is introduced into the network, the model can more fully capture spectral correlations, with accuracy improvements of 3.11%, 1.77%, 1.42%, and 0.5%, respectively. Furthermore, with the addition of SPE-LGAG, the model can focus more on key regions and result in a slight improvement in performance.

After adding ARSM to the baseline, the model can capture global spatial context more efficiently in sequence modeling by structuring spatial features, resulting in an improvement of 4.65%, 3.41%, 2.32%, and 0.77% in accuracy, respectively. On this basis, further introduction of SPA-LGAG also brings slight performance improvement. Ultimately, when the model integrates all modules simultaneously, it can achieve the optimal classification performance in all experimental settings.

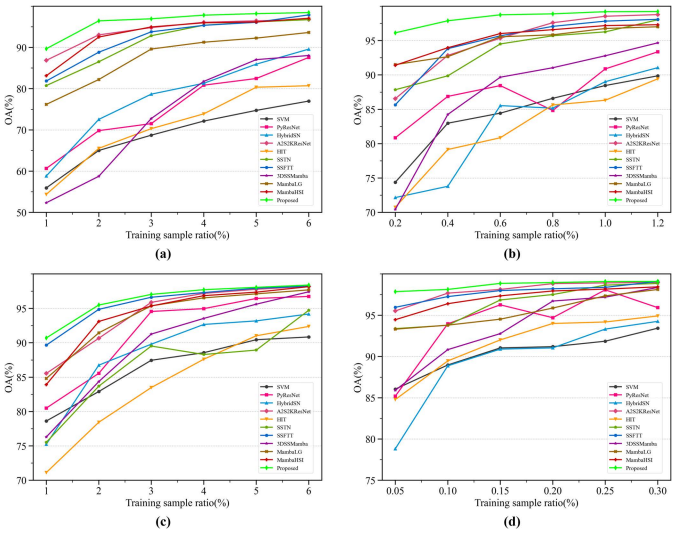


Fig. 11. OA values of all methods under different training samples. (a)–(d) IPs, PU, HST, and LK.

D. Different Sample Proportions

To evaluate the robustness of ARGO-Mamba, this article conducted experiments using different numbers of training samples on each dataset, and the experimental results are shown in Fig. 11. It can be seen that PyResNet showed significant fluctuations on the PU and LK datasets, while SSTN showed significant fluctuations on the HST dataset, indicating limitations in feature extraction and utilization capabilities. On

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the contrary, ARGO-Mamba maintains stable and excellent performance in both cases of sufficient and limited samples, demonstrating stronger feature learning and generalization abilities.

E. Model Complexity Analysis

To evaluate the practical efficiency of ARGO-Mamba, this section conducts a comparative analysis from three aspects: parameter count (Params), computational complexity (FLOPs), and inference time. As shown in Table VIII, the parameter count of the proposed model remains stable at around 0.4 M, significantly outperforming heavyweight architectures such as HIT (>48 M) and PyResNet (>21 M). Although models like SSTN are more compact, the quantitative experimental results combined with various datasets indicate that ARGO-Mamba achieves significant accuracy improvement (e.g., achieving 96.94% OA on the IP dataset, far higher than SSTN's 92.85%) with minimal additional overhead, achieving an optimal tradeoff between classification performance and hardware resources. Furthermore, the model's inference speed increases reasonably with the scale of data, demonstrating good scalability and low-latency advantages. In summary, ARGO-Mamba is a highly competitive lightweight, high-performance hyperspectral classification solution.

V. Conclusion

This article proposes an ARGO-Mamba model for HSIC tasks, which collaboratively models through spatial and spectral branches, effectively utilizing the complementary advantages of hyperspectral data. In the ARSM module, directional spatial encoding is utilized to replace traditional scanning optimization strategies, significantly enhancing the spatial modeling ability and classification performance of the model. At the same time, the GOSM module generates multiple views through a group offset mechanism, modeling spectral dependencies from different perspectives and reducing the impact of local dependency fragmentation caused by grouping. The experimental results indicate that ARGO-Mamba outperforms existing state-of-the-art methods in HSIC tasks.

Although the method proposed in this article has good classification performance, there are still several issues worth further study. First, how to more effectively preserve spatial semantic associations in sequence modeling remains a key challenge. Second, there is still room for improvement in the dynamic collaborative modeling of spatial and spectral features. Future work will be based on the proposed ARGO-Mamba framework to further investigate lightweight and low-power model design and expand to more challenging task scenarios, such as semisupervised classification, verifying the generalization and robustness of the model in practical applications.

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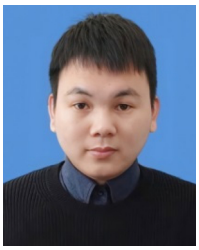
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